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Chapter 1

Materials and Methods

This chapter describes the experimental apparatus used in this work. First, a brief overview of the implementation of the Magneto-Optical Trap (MOT) for rubidium-87 atoms (^{87}Rb) is presented (this setup was already available prior to the present work). The chapter then focuses on the work carried out during this thesis, namely the implementation of the Optical Dipole Trap setup employed to transfer and trap atoms from the MOT. A particular focus is put on the coupling between the ODT laser source and the Hollow-Core Photonic Crystal Fiber (HCPCF), that delivers the ODT light in the vacuum chamber.

This section describes the apparatus used to produce, cool, and trap a sample of ^{87}Rb atoms. Preparing the atomic sample relies on a well-established two-stage sequence: a Magneto-Optical Trap (MOT) stage, followed by an optical molasses phase. Together, these stages deliver a cold atomic cloud, which constitutes the starting point for loading the Optical Dipole Trap (ODT) — the main addition developed during this thesis work.

Spatial confinement and initial cooling of the background rubidium vapor are provided by the MOT, which relies on the scattering force generated by three pairs of counter-propagating, red-detuned laser beams combined with a quadrupole magnetic field gradient [1]. In each pair the two beams carry opposite circular polarization, so that atoms moving toward either beam preferentially absorb photons from the beam pushing them back toward the trap center, while the quadrupole field makes this restoring effect position-dependent through the atomic Zeeman shift [1]. This stage compresses and Doppler-cools the atomic cloud. Once sufficient density and low enough temperature are reached, the magnetic field gradient is switched off and the atoms are left in the light field alone, in an optical molasses configuration; the $\sigma^+ - \sigma^-$ polarization arrangement of the MOT beams then drives further, sub-Doppler cooling [1, 2].

1.0.1 Setup Description

At the heart of the apparatus is a vacuum chamber operated under Ultra-High Vacuum (UHV) conditions, in which the atoms are cooled and trapped; the chamber is rigidly clamped to an optical table. Its overall layout and the surrounding experimental elements are shown in the accompanying CAD rendering.

UHV conditions are reached in two steps. An external dual diaphragm-turbo pumping station (Pfeiffer Vacuum HiCube 80 Eco DN63) first brings the chamber down to roughly 10^{-7} mbar; at that point, an internal combination pump (SAES Getter NEX Torr[®] Z200), consisting of an ion pump and a non-evaporable getter (NEG), is switched on and maintains a steady-state pressure around 10^{-10} mbar, as monitored through the ion-pump readout.

Optical access to the chamber is provided by eight windows. Two large ones (CF100 flanges, ~ 100 mm diameter) sit along the z axis, while two further pairs of smaller windows (CF40 flanges, ~ 40 mm diameter) are tilted at 45° with respect to the table; all six of these carry a 780 nm anti-reflective coating to optimize transmission of the MOT beams. The two remaining CF40 ports use uncoated glass: one on top of the chamber (along $+y$), used to inject the second ODT beam, and one along $+x$ on a 4-way cross, originally meant for imaging. In practice this x -axis view looks directly onto the ion-pump face, so the associated reflections introduce a strong parasitic background; imaging was therefore instead performed from the two large z -axis windows, which give a much cleaner background.

The rubidium vapor is supplied by a 1 g ampoule of pure ^{87}Rb , connected via a valve to one port of the 4-way cross (the fourth port connects to the external pump), and heated with a variac-driven heating band. Opening the rubidium valve raises the chamber pressure from $\sim 10^{-10}$ mbar to $\sim 10^{-8}$ mbar.

Trapping requires a spatially varying magnetic field to Zeeman-shift the atomic levels position-dependently, which in turn produces the position-dependent radiation pressure that pushes atoms toward the field zero. This quadrupole field is generated by a pair of circular coils (110 mm inner diameter) on the $+z/-z$ sides of the chamber, wired in anti-Helmholtz configuration. The field vanishes at the chamber center and grows linearly nearby, with a measured axial gradient of $1.33 \text{ G cm}^{-1} \text{ A}^{-1}$; since $\nabla \cdot \mathbf{B} = 0$, the radial gradient is half this value with opposite sign — a consequence of the same divergence-free argument that fixes the ratio between axial and radial gradient in any coil pair of this kind [3] — which is what gives the MOT cloud its characteristic ellipsoidal shape.

Three further pairs of square compensation (“shim”) coils (~ 650 mm side, mutually orthogonal along x , y , z) are driven in Helmholtz configuration to produce a uniform, arbitrarily oriented field at the chamber center. Beyond cancelling stray fields such as the Earth’s, these coils were also used to displace the MOT position and improve its spatial

overlap with the ODT beam. The field produced scales linearly with the applied current, with calibration factors [2]

$$B_z \approx 1.4 \text{ mG/mA}, \quad B_x \approx 1.3 \text{ mG/mA}, \quad B_y \approx 1.5 \text{ mG/mA}. \quad (1.1)$$

The bottom of the chamber (along $-y$) carries a CF40 blind flange fitted with a 1/8" Swagelok feedthrough; a Teflon ferrule with a 0.5 mm bore hosts the Hollow-Core Photonic Crystal Fiber (HCPCF) while preserving the vacuum seal. Inside the chamber, a custom mount holds the fiber in strict alignment, with its tip roughly 10 mm below the chamber center.

Six MOT beams are launched into the chamber via three fiber collimators (Schäfter+Kirchhoff, 60FC-Q780-4-M100S-37) and retro-reflected by external mirrors; a quarter-wave plate placed in front of each mirror is traversed twice by the light, giving the two counter-propagating beams of each pair opposite circular polarization (σ^+/σ^-), as required for MOT operation and sub-Doppler cooling. The retro-reflected beams carry about 88% of the incident power; although not perfectly balanced, this asymmetry is small enough that trapping is not noticeably affected [4].

Aside from the master-laser locking branch, the ODT injection path built for this thesis, and the MOT beams themselves, the rest of the apparatus is entirely fiber-coupled, using polarization-maintaining, absorption-reducing (PANDA) fiber [1]. In this design two stress-applying rods flank the core and mechanically induce a strong, fixed birefringence between two orthogonal axes; provided the input polarization is aligned with one of these axes and the a fiber is kept reasonably free of twists, this birefringence suppresses coupling between the two polarization eigenmodes and keeps the output polarization stable [1]. Beyond this polarization benefit, the fully fiber-coupled design limits misalignment from thermal drifts, keeps the optical layout compact, and reduces the safety hazards of free-space high-power beams — advantages that also matter for possible future replication of the setup on other experimental platforms.

1.0.2 Laser System and Frequency Stabilization

Cooling operates on the $5^2S_{1/2} \rightarrow 5^2P_{3/2}$ (D2) transition of ^{87}Rb near 780 nm. A full description of the cooling scheme requires the hyperfine structure, labeled by total angular momentum $F = I + J$ (I : nuclear spin, J : total electronic angular momentum); the relevant hyperfine levels and laser-coupled transitions for both ^{85}Rb and ^{87}Rb are shown in the accompanying level diagram.

The cooling light drives $F = 2 \rightarrow F' = 3$. Before the MOT light is switched on, thermal ^{87}Rb atoms are split between the $F = 1$ and $F = 2$ ground states in a statistical-weight ratio of 3 : 5. Once cooling starts, atoms are excited out of $F = 2$ toward $F' = 3$; since $\Delta F = 0, \pm 1$, the only allowed decay channel is back to $F = 2$, making this a closed

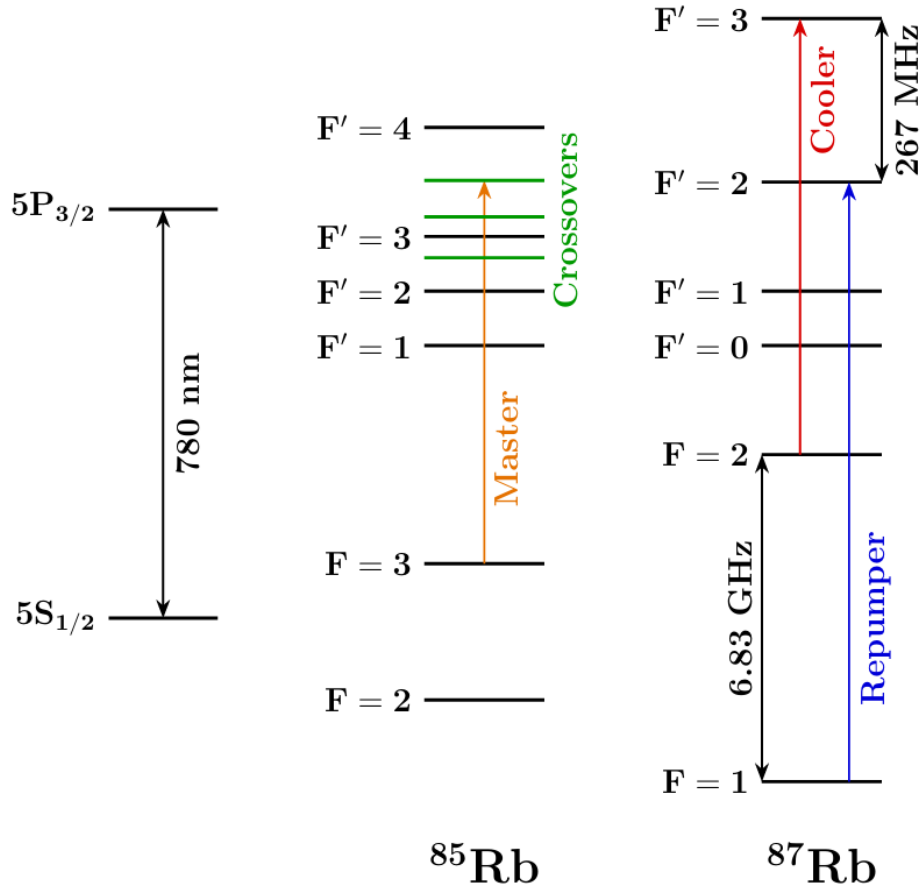


Figure 1.1: *Level scheme of the 85Rb and 87Rb D2 line.*

transition in the ideal case. A small off-resonant excitation path, $F = 2 \rightarrow F' = 2$, exists as well, from which atoms can decay either back to $F = 2$ or into the dark state $F = 1$, where they no longer interact with the cooling light. A repumping laser resonant with $F = 1 \rightarrow F' = 2$ returns these atoms to the cooling cycle (and simultaneously recovers the thermal population initially sitting in $F = 1$), which improves the overall MOT loading efficiency.

Both cooling and repumping light must be tightly frequency-locked, with an appropriate detuning, to their respective transitions; the locking schemes are outlined below, with a more complete treatment available in [5, 4].

1.0.2.1 Master laser

Rather than being locked directly, the cooler and repumper lasers are referenced to a third, master laser, which is itself frequency-stabilized via saturation spectroscopy [1]. In this technique two counter-propagating beams derived from the same source — an intense “pump” and a weak “probe” — cross inside a heated vapor cell; because only atoms with a near-zero velocity component along the beam axis are simultaneously resonant with both

beams, the pump partially saturates the transition and burns a narrow, Doppler-free feature into the probe absorption profile at line center, which is what allows the laser to be locked precisely on resonance rather than merely somewhere inside the much broader Doppler-limited line [1]. The master frequency is conveniently chosen between the cooling and repumping frequencies, and is locked to the $(F = 3 \rightarrow F' = 3) \times (F = 3 \rightarrow F' = 4)$ crossover of ^{85}Rb .

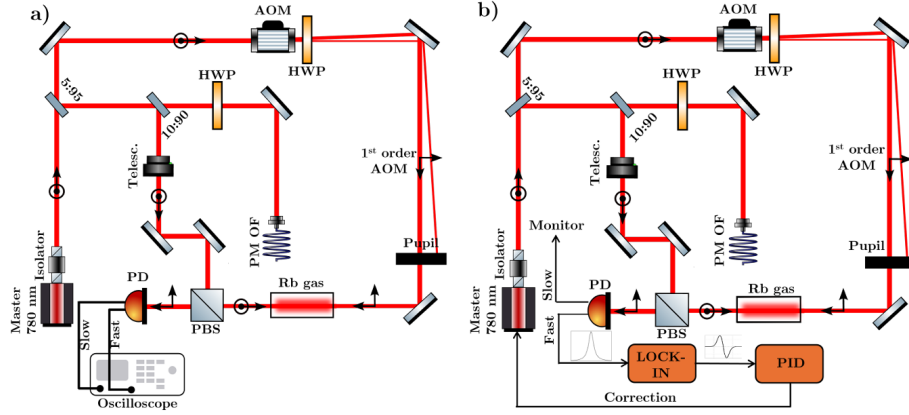


Figure 1.2: *Master laser locking setup scheme.*

The light source is a 780 nm current-driven, temperature-stabilized DFB laser diode (Topica EYP-DFB-0780-00040-1500-BFW11-0005). A 5:95 beam splitter separates a weak "probe" branch (5%) from the main branch (95%), which is further split 10:90: the larger fraction feeds the cooler/repumper locking setup, while the remaining 10% — the "pump" beam — is used for saturation spectroscopy.

The pump beam is fully reflected off a polarizing beam splitter (PBS) and sent through a heated ^{85}Rb vapor cell. The probe beam, on the other branch, first passes through an acousto-optic modulator (AOM) driven by an amplified RF tone from a Direct Digital Synthesizer (DDS) under experiment control; only the first diffraction order is kept, the zeroth order being blocked by an iris. The AOM frequency-modulates the probe beam as needed for the locking protocol, so that the transmitted signal can later be demodulated into a dispersive, zero-crossing error signal centered on the atomic resonance rather than a featureless absorption dip. After a half-wave plate sets its polarization, the probe beam is sent counter-propagating through the same vapor cell, then transmitted through the PBS onto a photodiode. The photodiode provides both a slow output, monitored on an oscilloscope, and a fast output, which is demodulated to extract the dispersive error signal; this signal drives a PID controller that feeds back onto the laser diode current to keep the master frequency locked.

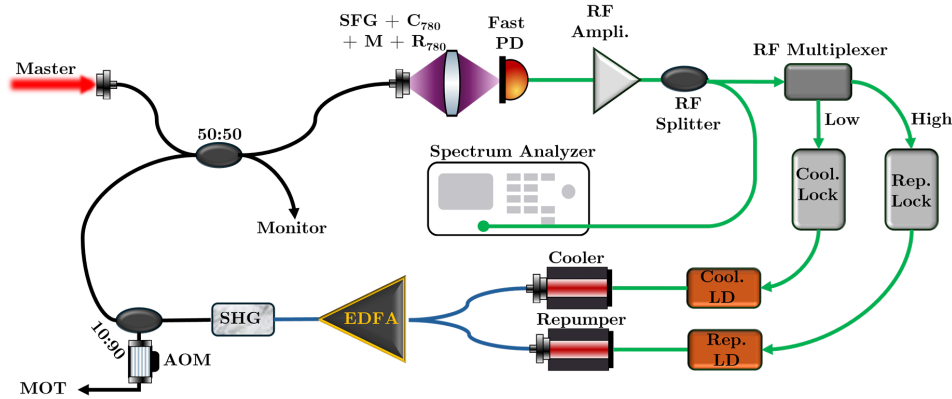


Figure 1.3: *Beat-note locking setup scheme.*

1.0.2.2 Cooler and repumper lasers

Cooling and repumping light originate from two 1560 nm DFB diodes (Eblana EP1562-0-NLW-B26-100FM), fiber-coupled and combined into a single fiber before being amplified by an Erbium-Doped Fiber Amplifier (EDFA); under normal operation the EDFA is driven with ~ 5.2 mW of input power and delivers ~ 500 mW at its output, per its built-in monitor.

The amplified 1560 nm light is then frequency-doubled to 780 nm in a nonlinear crystal via Second Harmonic Generation (SHG); the same crystal also produces an unused sum-frequency component (SFG) from the mixing of the cooler and repumper fields. After the SHG stage, a 10:90 beam splitter sends 90% of the light through a fast-switching AOM to the three MOT beam collimators, while the remaining 10% is combined, via a 2×2 fiber splitter, with the master-laser light to feed the cooler/repumper locking chain. (During this thesis work the original 10:90 splitter failed and was temporarily substituted with a 25:75 unit.)

The combined optical field — master, cooler, repumper, and SFG components together — is detected on a fast, ~ 10 GHz-bandwidth photodiode. Beating between these frequency components produces amplitude modulation at the corresponding beat frequencies; the two of interest are the master-cooler beat near 1 GHz and the master-repumper beat near 5.5 GHz. After amplification, the photodiode signal is split by RF filters into low- and high-frequency bands to isolate these two beats, each of which feeds its own PID controller driving the corresponding laser diode current. In this scheme — known as frequency offset locking, or beat-note locking — it is the frequency difference between each secondary laser and the master that is stabilized, rather than the absolute frequency of the secondary laser itself; this makes it straightforward to keep the cooler and repumper at a fixed, precisely known offset from the master reference regardless of slow drifts common to the whole system [3].

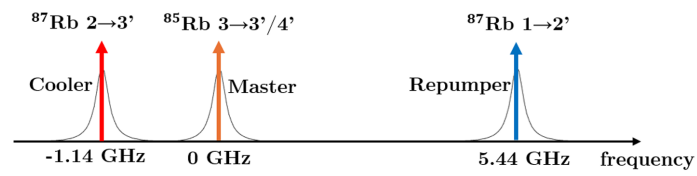


Figure 1.4: *Example of beat-note signals on a spectrum analyzer, used for frequency offset locking of the cooler and repumper lasers relative to the master laser.*

1.1 Experimental Procedures and Analysis Methodology

Every experimental run begins with loading atoms into the MOT, using the operating point already established for this apparatus [1]. The trap is operated with an axial magnetic field gradient of $B' = 9.32 \text{ G/cm}$ (an 8 A coil current), a cooler detuning of $\Delta = -13 \text{ MHz}$, and all compensation coils set to zero current; the MOT is held for 5 s before proceeding.

Loading is followed by an optical molasses stage: the quadrupole field is switched off, which lets the temperature drop further but also removes the spatial confinement, so the cloud simultaneously expands and starts to fall under gravity while still damped by the light field. To first approximation the molasses temperature scales as $I/|\Delta|$, where I is the cooling-beam intensity and Δ the detuning [6] — a scaling that only holds as long as $|\Delta|$ stays small enough not to substantially suppress the photon scattering rate $\Gamma_p \propto 1/\Delta^2$, and not so large that the neighboring $F = 2 \rightarrow F' = 2$ transition starts being appreciably excited. Because molasses cooling also works best with a genuinely zero magnetic field, the shim coil currents are updated 5 ms after the MOT field is switched off to null the Earth’s field and any other residual field in the lab. The cooling-light detuning itself is changed from its MOT value to the molasses value through a 20-step ramp lasting 5 ms in total, starting 50 μs after the shim-coil currents have been updated.

1.1.1 Atom Imaging

Two monochromatic CMOS cameras were used to image the atomic cloud inside the chamber:

- a Teledyne FLIR BFS-U3-04S2M-CS, based on the Sony IMX287 sensor, with a 720×540 resolution and $6.9 \mu\text{m} \times 6.9 \mu\text{m}$ pixels;
- an Allied Vision Manta G-419 NIR, based on the CMOSIS/ams CMV4000 sensor, with a 2048×2048 resolution and $5.5 \mu\text{m} \times 5.5 \mu\text{m}$ pixels.

The FLIR camera was used mainly while optimizing the molasses stage, while the Manta camera—chosen for its better sensitivity at 780 nm and finer pixel pitch, well suited to resolving the small region occupied by the ODT—was used to measure the ODT capture efficiency. Both cameras were fitted with a Thorlabs MVL50M23 adjustable objective (50 mm focal length, 200 mm minimum working distance, giving a minimum magnification of ~ 0.33), adjusted so that the HCPCF inside the chamber—and, at essentially the same distance, the atomic cloud—was in focus. Both cameras were triggered externally by the FPGA control system.

1.1.2 Camera Calibration

Accurate temperature measurements of the atomic cloud rely on the precise determination of its spatial dimensions. Consequently, a rigorous spatial calibration of the imaging system (FLIR BlackFly camera) is required to establish the effective pixel size in the object plane. Given the physical pixel size of the camera sensor ($6.9\ \mu\text{m}$), four distinct methodologies were evaluated to determine the optical magnification M and the corresponding effective pixel size.

1.1.2.1 Method 1: Direct Imaging of a Known Microstructure

The first calibration approach utilized a Hollow-Core Photonic Crystal Fiber (HCPCF) with a precisely known physical diameter of $316\ \mu\text{m}$ as a spatial reference. The fiber was placed in the object plane and imaged. Due to optical blurring and the inherent difficulty in precisely identifying the boundary pixels of the fiber, a spatial fitting routine was employed to extract the apparent diameter.

The fit indicated that the $316\ \mu\text{m}$ diameter corresponded to an extent of 8 pixels on the sensor. This yields an effective pixel size in the object space of:

$$C = \frac{316\ \mu\text{m}}{8\ \text{px}} = 39.5\ \mu\text{m}/\text{px} \quad (1.2)$$

which corresponds to an optical magnification of $M = 6.9/39.5 = 0.174$.

1.1.2.2 Method 2: Kinematic Tracking of Free Fall

A dynamic calibration was performed by tracking the free-fall trajectory of the atomic cloud's center of mass under the influence of gravity. By analyzing the vertical displacement as a function of the time-of-flight, the spatial scale can be deduced.

This kinematic approach yielded an effective pixel size of $50.6\ \mu\text{m}/\text{px}$, corresponding to a magnification of $M = 0.136$. However, this method is susceptible to systematic errors and requires further refinement. Specifically, the spatial non-uniformity of the probe beam intensity profile skews the apparent center of mass, as the detected optical signal is directly proportional to the local atomic fluorescence or absorption.

1.1.2.3 Method 3: Optical Target Calibration via Reflection

To ensure the calibration target was positioned exactly at the focal plane corresponding to the atomic cloud, a macroscopic optical target (a standard ruler) was utilized in conjunction with a mirror. The camera, already focused on the atoms from previous experimental runs, imaged the reflection of the ruler. The ruler was carefully positioned until its reflected image was perfectly sharp, guaranteeing optical equidistance.

Multiple segments of the ruler were analyzed to provide a statistically robust measurement:

- A 2 cm segment corresponded to 471 px ($42.46 \mu\text{m}/\text{px}$).
- A 2.5 cm segment corresponded to 571 px ($43.78 \mu\text{m}/\text{px}$).
- A 1-inch segment (2.54 cm) corresponded to 587 px ($43.27 \mu\text{m}/\text{px}$).
- A 3 cm segment corresponded to 685 px ($43.79 \mu\text{m}/\text{px}$).

Averaging these results yields a highly consistent effective pixel size of approximately $43.3 \mu\text{m}/\text{px}$, corresponding to a magnification of $M \approx 0.160$.

1.1.2.4 Method 4: Magnetic Field Displacement

The final calibration technique compared the theoretically predicted spatial displacement of the Magneto-Optical Trap (MOT) against the experimentally observed displacement in pixels when a known bias magnetic field was applied.

Experimentally, the application of a uniform bias field via the y -axis shim coils resulted in a measured spatial shift of $y_{\text{shift}} = 58 \text{ px/A}$. Analytically, the shim coils—approximated as a cubic structure with side length $L = 645 \text{ mm}$ and $N = 100$ loops—provide a magnetic field transfer function of $b = 1.43 \text{ G/A}$. The restoring force of the MOT is governed by its quadrupole magnetic field gradient. The MOT coils, operating at a current of $I = 8.4 \text{ A}$ with 128 loops, an effective radius $R_{\text{eff}} = 62 \text{ mm}$, and an effective separation $a_{\text{eff}} = 162 \text{ mm}$, produce a calculated vertical gradient of $G_{y\text{TOT}} = 0.571 \text{ G/mm}$.

The effective pixel size C is derived by equating the magnetic displacement to the pixel displacement:

$$C = \frac{b}{G_{y\text{TOT}} \cdot y_{\text{shift}}} = \frac{1.43 \text{ G/A}}{0.571 \text{ G/mm} \cdot 58 \text{ px/A}} \approx 0.043 \text{ mm/px} = 43 \mu\text{m/px} \quad (1.3)$$

Alternatively, utilizing empirically derived values from previous characterizations of the apparatus ($b = 1.5 \text{ G/A}$ and $G = 5.6 \text{ G/cm}$ at 8.4 A), the calibration yields:

$$C = \frac{1.5}{5.6 \cdot 58} \text{ cm/px} \approx 46 \mu\text{m/px} \quad (1.4)$$

These magnetic displacement results show excellent agreement with the optical target calibration described in Method 3.

1.1.3 Temperature Measurements

Temperature measurements were used to guide the optimization of the molasses parameters toward the lowest achievable temperature. The method relies on the ballistic ex-

pansion of the atomic cloud during free fall. For a non-interacting gas, along a single coordinate, an atom found at x_0 at $t = 0$ is at position

$$x(t) = x_0 + v_x t \quad (1.5)$$

after a free evolution time t . Treating position and velocity as statistically independent random variables, the variance of their sum adds, so that

$$\sigma_x^2(t) = \sigma_{x,0}^2 + \sigma_{v_x}^2 t^2, \quad (1.6)$$

with $\sigma_{x,0}$ the initial position spread. The equipartition theorem relates the mean squared velocity along x to the corresponding temperature,

$$\frac{1}{2}m\langle v_x^2 \rangle = \frac{1}{2}k_B T_x, \quad (1.7)$$

and, working in the center-of-mass frame of the cloud where $\langle v_x \rangle = 0$,

$$\langle v_x^2 \rangle = \sigma_{v_x}^2 = \frac{k_B T_x}{m}. \quad (1.8)$$

Combining these gives the expansion law actually used for the fit,

$$\sigma_x^2(t) = \sigma_{x,0}^2 + \frac{k_B T_x}{m} t^2, \quad (1.9)$$

which, using the ^{87}Rb atomic mass $m = m_A(^{87}\text{Rb}) = 86.9 \text{ u} \approx 1.443 \times 10^{-25} \text{ kg}$ and images acquired at a series of times of flight (TOF), yields T_x as a fit parameter.

The full acquisition and analysis sequence is as follows:

1. A MOT is loaded and followed by the molasses stage, whose parameters and duration are the ones being optimized (see the corresponding results chapter).
2. All cooling light is then switched off, letting the cloud fall and expand freely. The FLIR camera records an image $I(\text{TOF})$ at each chosen time of flight, together with a background image $I_{\text{bkg}}(\text{TOF})$ taken after waiting long enough (1 s) for the cloud to have fully dispersed. Because the imaging flash itself heats and destroys the sample, each TOF requires its own independent MOT-loading and molasses sequence.
3. The background is subtracted,

$$I_{\text{clean}}(\text{TOF}) = I(\text{TOF}) - I_{\text{bkg}}(\text{TOF}), \quad (1.10)$$

with an associated Poisson uncertainty

$$\sigma_N = \sqrt{I(\text{TOF}) + I_{\text{bkg}}(\text{TOF})}. \quad (1.11)$$

4. Each background-subtracted image is fitted with a two-dimensional Gaussian (via a trust-region-reflective minimization) to extract the cloud widths σ_x and σ_y along the two principal axes.
5. The expansion law is fitted independently along x and y to obtain T_x and T_y , and the reported temperature T is their average.

1.1.4 ODT Loading Measurements

Introduction

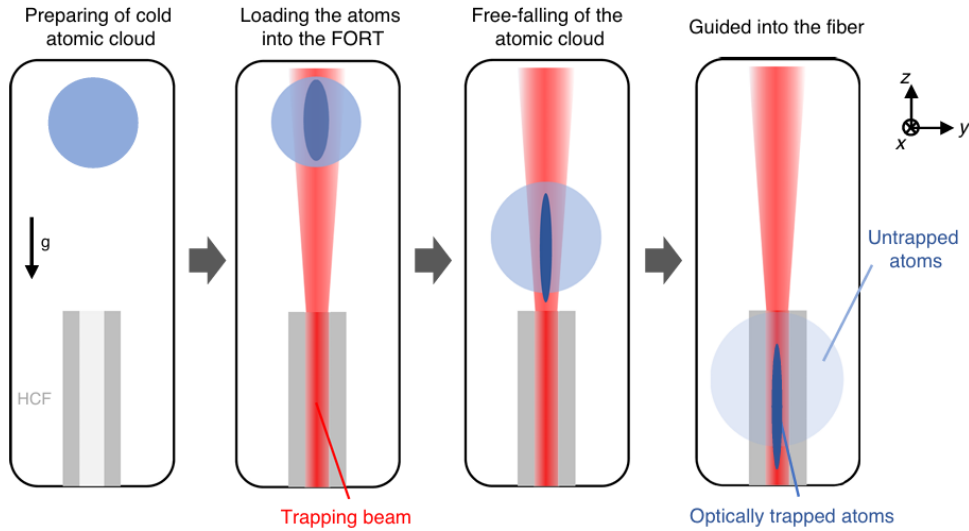


Figure 1.5: *Schematic of the differential imaging technique used for ODT loading measurements, comparing a sequence with the ODT on to one with the ODT off after a time-of-flight.*

The ODT performance is quantified by two figures of merit: the loading efficiency η and the integrated intensity of the ODT peak, which is directly proportional to the number of captured atoms. Both are extracted by comparing fluorescence images taken with the ODT switched on or off during the MOT and molasses stages. Since only a small fraction of the atoms — of order $1/1000$ — end up captured by the ODT, this comparison relies on differential imaging with careful background normalization, so that the much larger and slowly-fluctuating molasses signal does not swamp the small ODT signal. The full procedure is as follows.

1. Two otherwise identical MOT-plus-molasses sequences are run back to back, one with the ODT held on throughout (starting from the MOT loading phase) and one with it off. After a 30 ms TOF — during which the untrapped atoms expand and thereby become easier to distinguish from those confined by the ODT — fluorescence images I_{on} and I_{off} are recorded with the Manta camera. Each on/off pair is repeated five times.

2. Both images are cropped to a 530×380 pixel region of interest centered on the atomic cloud, which excludes both the region where the HCPCF itself is visible and peripheral areas containing only background.
3. To improve the signal-to-noise ratio — important given how weak the ODT signal is — the cropped images for each condition are averaged, giving $\bar{I}_{\text{on}} = \langle I_{\text{on}} \rangle$ and $\bar{I}_{\text{off}} = \langle I_{\text{off}} \rangle$.
4. These averaged images are then collapsed into one-dimensional density profiles by integrating along columns, which further improves the signal-to-noise ratio.
5. The total molasses atom number drifts slowly over time, even at fixed experimental parameters, due to slow pressure changes in the chamber and thermal effects in optical components such as the AOMs. Because these drifts are correlated between the on/off measurements and the ODT signal itself is small, simply differencing the averaged profiles leaves a residual background comparable in size to the ODT signal. To correct for this, the off profile is rescaled by a factor k chosen so that the integrated difference outside the ODT region vanishes:

- (a) An exclusion window $x_{\text{ODT}} \pm R_{\text{window}}$ around the ODT position is defined; everything outside it is the “outside region”.
- (b) A linear background $L_i(x)$ is estimated from the edges of each profile; after subtracting it, the integrated signal outside the ODT region is computed for both conditions,

$$A_i^{\text{out}} = \int_{\text{outside}} [P_i(x) - \bar{I}_i(x)] dx, \quad i \in \{\text{on}, \text{off}\}. \quad (1.12)$$

- (c) The rescaling factor is the ratio of these two areas,

$$k = \frac{A_{\text{on}}^{\text{out}}}{A_{\text{off}}^{\text{out}}}. \quad (1.13)$$

6. The rescaled differential profile, isolating the ODT-trapped atom signal, is then

$$\Delta \bar{I}(x) = \bar{I}_{\text{on}}(x) - k \cdot \bar{I}_{\text{off}}(x). \quad (1.14)$$

7. Two independent fits — each a Gaussian on top of a linear background — are performed: one on the off profile $\bar{I}_{\text{off}}(x)$ and one on the differential profile $\Delta \bar{I}(x)$. From the resulting Gaussian areas $A_{\text{off}}^{\text{Gaus}}$ and $A_{\text{ODT}}^{\text{Gaus}}$, the capture efficiency follows as

$$\eta = \frac{A_{\text{ODT}}^{\text{Gaus}}}{k \times A_{\text{off}}^{\text{Gaus}}}. \quad (1.15)$$

1.2 Measurement of number of atoms

This section details the experimental procedure and data analysis workflow for calibrating the Manta scientific camera, which is utilized to measure the absolute number of trapped atoms in an Optical Dipole Trap (ODT). The optical attenuation measurements, the derivation of the camera’s calibration factor—mapping photon counts to physical energy—and the subsequent extraction of the absolute atom number from fluorescence images are rigorously described.

1.2.1 Experimental Procedure

The experimental sequence initiates with a spherical Magneto-Optical Trap (MOT) capturing approximately 10^8 atoms at a temperature of $8\,\mu\text{K}$. The initial atomic cloud radius is approximately 1 mm, positioned 7 mm from the fiber. The atoms are subsequently loaded into a free-fall configuration Optical Dipole Trap (ODT) operating at an optical power of 0.8 W.

To extract the atomic spatial distribution, fluorescence imaging is employed. Two sequential images are acquired:

1. A signal image with both the molasses beams and the ODT active.
2. A background image with the molasses active but the ODT inactive.

Subtracting these two images yields a two-dimensional spatial profile comprising two Gaussian components. A flattening correction is applied to the broader background Gaussian to isolate the narrower Gaussian peak, which corresponds to the atomic distribution confined by the ODT. The total integrated area of this residual Gaussian, expressed in digital counts, is directly proportional to the number of trapped atoms.

1.2.2 Optical Attenuation and Power Calibration

To convert the integrated counts into a physical atom number, the imaging camera must be calibrated against a precisely known incident optical power. A calibration beam, generated via an Acousto-Optic Modulator (AOM), is directed towards the camera sensor. The calibration beam propagates through a series of optical attenuators and the vacuum chamber viewports. To accurately determine the optical power reaching the camera sensor (P_{cam}), the transmission coefficient of each optical element must be calculated. A constant background power of $P_{bg} = 1.5\,\mu\text{W}$ is systematically subtracted from all radiometric readings. The initial power prior to the first attenuator (at an AOM amplitude setting of 450) is measured to be $1000\,\mu\text{W}$.

The transmission factors (T) for the individual components are computed as follows:

$$T_1 = \frac{182 - 1.5}{1000 - 1.5} = \frac{180.5}{998.5} \approx 0.1808 \quad (\text{Attenuator 1}) \quad (1.16)$$

$$T_{vac} = \frac{870 - 1.5}{1000 - 1.5} = \frac{868.5}{998.5} \approx 0.8698 \quad (\text{Vacuum Chamber}) \quad (1.17)$$

$$T_2 = \frac{540 - 1.5}{950 - 1.5} = \frac{538.5}{948.5} \approx 0.5677 \quad (\text{Attenuator 2}) \quad (1.18)$$

$$T_3 = \frac{78 - 1.5}{540 - 1.5} = \frac{76.5}{538.5} \approx 0.1420 \quad (\text{Attenuator 3}) \quad (1.19)$$

The total transmission of the system is the product of these individual factors:

$$T_{tot} = T_1 \cdot T_{vac} \cdot T_2 \cdot T_3 \approx 0.01272 \quad (1.20)$$

During the calibration exposure sequence, the AOM amplitude is set to 50, providing an initial measured power of $22 \mu\text{W}$ (after background subtraction). The effective power reaching the camera is therefore:

$$P_{cam} = 22 \mu\text{W} \cdot T_{tot} \approx 279.8 \text{ nW} \quad (1.21)$$

1.2.3 Camera Calibration Factor (K)

To determine the calibration factor K (in photons per count), the camera's exposure time (t_{exp}) is varied while maintaining a constant incident power P_{cam} . Let the total counts extracted from the image be denoted as C . Assuming a linear response from the sensor, the counts accumulate linearly as a function of the exposure time:

$$C = m \cdot t_{exp} + q \quad (1.22)$$

By performing a linear regression on the experimental data, the slope m (in counts per second) is extracted. The rate of incident photons is given by the optical power divided by the energy of a single photon ($E_{ph} = hc/\lambda$). Therefore, the calibration factor K is analytically defined as:

$$K = \frac{\text{Incident Photons}}{\text{Total Counts}} = \frac{P_{cam} \cdot \lambda}{m \cdot h \cdot c} \quad (1.23)$$

where h is Planck's constant, c is the speed of light, and λ is the laser wavelength.

1.2.4 Atom Number Calculation

Once the camera is calibrated, the integrated area of the residual Gaussian from the ODT image (C_{tot} , expressed in counts) is converted into the total number of trapped atoms (N).

First, the total number of detected photons is calculated as:

$$N_{det} = C_{tot} \cdot K \quad (1.24)$$

During the imaging pulse duration (t_{img}), each atom scatters light isotropically at a rate R_{sc} , governed by the two-level atom scattering formula:

$$R_{sc} = \frac{\Gamma}{2} \frac{I/I_{sat}}{1 + I/I_{sat} + 4(\Delta/\Gamma)^2} \quad (1.25)$$

where Γ is the natural linewidth, I is the total imaging intensity, I_{sat} is the saturation intensity, and Δ is the laser detuning.

Only a specific fraction of these scattered photons reaches the camera sensor. This fraction is determined by the solid angle of the collection optics (Ω) and the optical transmission of the viewports and lenses (T_{optics}). The fractional solid angle is approximated as $\eta = r^2/(4d^2)$, where r is the effective radius of the collection lens and d is the distance from the atoms to the lens.

The number of photons arriving at the camera sensor per single atom is thus:

$$N_{ph/atom} = R_{sc} \cdot t_{img} \cdot \eta \cdot T_{optics} \quad (1.26)$$

Finally, the absolute number of trapped atoms is derived by dividing the total detected photons by the single-atom photon contribution:

$$N = \frac{N_{det}}{N_{ph/atom}} = \frac{C_{tot} \cdot K}{R_{sc} \cdot t_{img} \cdot \frac{r^2}{4d^2} \cdot T_{optics}} \quad (1.27)$$

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1.3 Injection into the Black Fiber

The goal of this stage of the work is to couple approximately 1 W of 1064 nm light into the so-called “black fiber” with the highest possible efficiency, as a preliminary step before characterizing its polarization stability. Bending losses were checked early on and found to be negligible, with less than 1% power reduction under moderate bending of the fiber.

1.3.1 Beam profiling procedure

Throughout this work, characterizing the size and divergence of a beam relied on a common beam-profiling procedure, applied both before and after the fiber whenever the injection geometry needed to be checked. The beam under test was first made horizontal and set at a fixed height on the optical table, so that it could be sampled at a sequence of positions along its propagation direction without introducing additional steering optics. A camera mounted on a translation stage was then used to record images of the beam cross-section at several distances along this direction.

Each recorded image was fitted with a two-dimensional Gaussian intensity profile to extract the beam radius along the two transverse axes at that particular distance, after appropriate cropping of the raw images to isolate the beam spot from background and stray light. Repeating this at multiple positions yielded the beam radius as a function of the longitudinal coordinate z , which was then fitted to the standard Gaussian beam propagation law

$$w(z) = w_0 \sqrt{1 + \left(\frac{z - z_0}{z_R} \right)^2}, \quad (1.28)$$

with w_0 the waist size, z_0 the waist position, and z_R the associated Rayleigh range. This fit provided, for any measurement point in the setup, the waist size and position of the beam, which could then be compared to the target mode parameters of the fiber or used to design the appropriate injection optics.

1.3.2 First injection attempts and the role of beam divergence

The first configuration tested consisted of a collimated beam sent directly into a fixed $f = 18$ mm fiber collimator (Schäfter+Kirchhoff, 60FC-4-A18-03) attached to the fiber. This yielded an injection efficiency of only about 20%. Tuning the divergence of the upstream beam expander so as to send a slightly diverging beam onto the collimator — roughly equivalent to inserting a weak $f = -1000$ mm lens — raised the efficiency

to above 50%. This was initially counterintuitive, since a fiber collimator is nominally designed to perform best with a collimated input.

Beam profiling before and after the fiber, following the procedure described above, clarified the origin of this behavior. The waist reconstructed from the post-fiber measurements agreed well with the mode field diameter specified by the manufacturer, confirming that the fundamental mode was being excited correctly. The need for input divergence was traced back to the presence of an endcap: a glass cone about 300 μm long spliced onto the fiber tip to improve power handling. The endcap effectively shifts the position of the fiber facet relative to the collimator's focal plane, so that a purely collimated input is no longer the optimal injection condition. Reproducing the required divergence with a dedicated lens would call for a long, weak $f \approx -1000\text{ mm}$ element placed immediately before the collimator — an impractical addition that would complicate the setup and likely degrade beam quality. This motivated replacing the fixed $f = 18\text{ mm}$ collimator with a more flexible arrangement: an $f = 25\text{ mm}$ lens mounted on a micrometric translation stage along the propagation axis, positioned about 100 μm closer to the fiber than would be expected without the endcap.

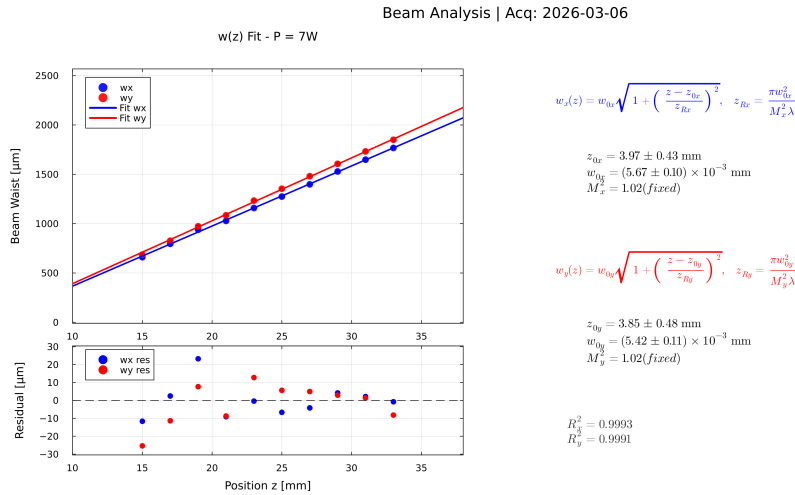


Figure 1.6

1.3.3 Damage threshold of the injection lens

The chosen lens (Thorlabs AC127-025-C) is a cemented achromatic doublet, and prolonged exposure to high optical power can degrade the cement layer. Its damage threshold was therefore checked against the manufacturer specification before proceeding: Thorlabs quotes a laser-induced damage threshold (LIDT) of 1000 W/cm at 1540 nm for this class of unmounted achromatic doublets.

Assuming a uniform beam profile, the average linear power density (LPD) for 1 W of power over a 0.32 cm beam diameter is

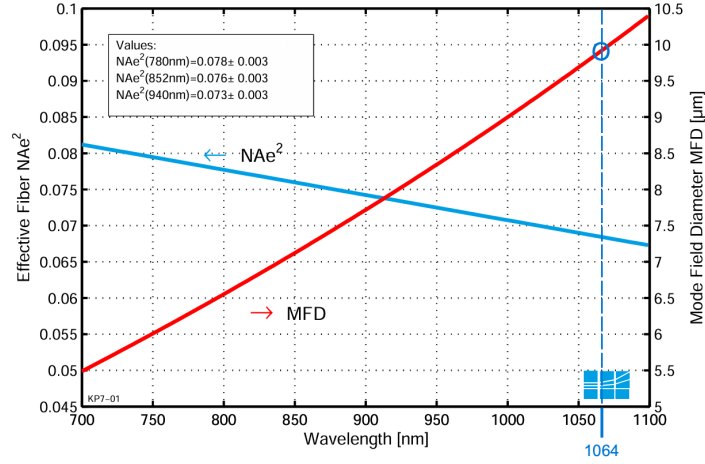


Figure 1.7

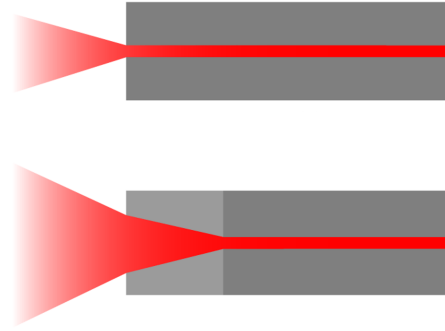


Figure 1.8

$$\text{LPD} = \frac{P}{D} = \frac{1.0 \text{ W}}{0.32 \text{ cm}} \approx 3.12 \text{ W/cm.} \quad (1.29)$$

A Gaussian beam reaches roughly twice this power density at its peak, giving

$$P_{\text{max}} \approx 2 \times \text{LPD} \approx 6.25 \text{ W/cm.} \quad (1.30)$$

Since the damage threshold scales approximately linearly with wavelength, the manufacturer's value can be rescaled from the 1540 nm test wavelength to the 1064 nm operating wavelength:

$$\text{LIDT}_{1064} = 1000 \text{ W/cm} \times \frac{1064 \text{ nm}}{1540 \text{ nm}} \approx 690.91 \text{ W/cm.} \quad (1.31)$$

Comparing the two figures,

$$6.25 \text{ W/cm} \ll 690.91 \text{ W/cm,} \quad (1.32)$$

confirms a large safety margin, so operation at the intended power was considered safe

for the lens.

1.3.4 Determining the required input beam size

With the $f = 25$ mm lens positioned roughly 25 mm from the fiber and a second collimator placed at the fiber output, the transmitted beam was profiled, again following the procedure above, to characterize the injection. Rather than emerging collimated, the beam was found to converge to a waist of about $280\text{ }\mu\text{m}$ at a distance of 1 m from the output — evidence of residual wavefront curvature and thus of a still sub-optimal mode match.

In principle, the input beam could have been shaped into the time-reversed copy of this output profile — a diverging beam with a radius of about $1200\text{ }\mu\text{m}$ on the collimator facet — but producing such a diverging beam reliably is experimentally difficult. Since the axial position of the $f = 25$ mm lens could instead be adjusted freely on its translation stage, the more practical route was to work out the collimated input beam radius w_0 needed for this lens to image the fiber mode correctly, using

$$w_0 = \frac{\lambda f}{\pi w'_0}, \quad (1.33)$$

where w'_0 is the waist at the fiber input determined from the earlier measurements. With $\lambda = 1.064\text{ }\mu\text{m}$, $f = 25\text{ mm}$, and $w'_0 = 5.2\text{ mm}$,

$$w_0 = \frac{(1.064 \times 10^{-6})(25 \times 10^{-3})}{\pi (5.2 \times 10^{-3})} \approx 1.6\text{ mm}. \quad (1.34)$$

The beam expander was then adjusted to produce a collimated beam with this radius, and the resulting beam was verified by profiling before proceeding to injection.

1.3.5 Diagnosing and correcting a low-efficiency injection

Injecting this properly-sized collimated beam through the $f = 25$ mm lens, however, initially gave a disappointing coupling efficiency of only about 1%, as measured with a power meter after the output collimator. To understand the discrepancy, the alignment was checked by sending light the other way through the system, directly through the fixed $f = 18$ mm collimator, while observing the beam from the $f = 25$ mm lens on a camera. This revealed that the lens was not producing a collimated output as intended: instead, the beam was focusing between the two steering mirrors, at a distance of about 125 mm from the lens.

Treating this focus as the effective object position allowed the correct lens position to be recovered from the imaging equation,

$$\frac{1}{p} + \frac{1}{q} = \frac{1}{f}, \quad (1.35)$$

with $q = 125$ mm and $f = 25$ mm, giving $p \approx 31.25$ mm. Since the lens must sit at its own focal distance from the fiber, $p = f = 25$ mm, in order to collimate the output correctly, the lens had to be moved closer to the fiber by

$$31.25 \text{ mm} - 25 \text{ mm} = 6.25 \text{ mm}. \quad (1.36)$$

After this 6 mm coarse correction, the lens position on the micrometric stage was fine-tuned until the beam transmitted through it was visibly collimated.

1.3.6 Final characterization

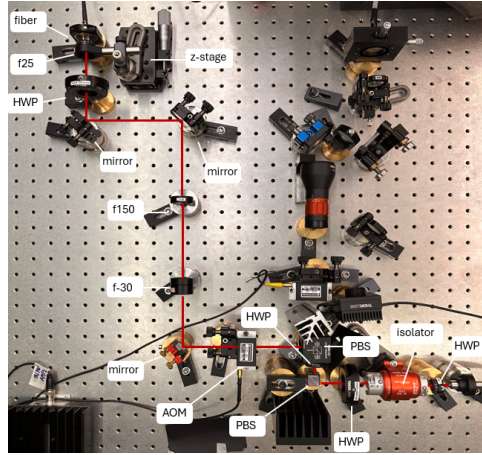
With the lens correctly positioned, the beam profile at 1064 nm and 7 W was measured after the black fiber, using the $f = 25$ mm lens as output collimator while injecting through the original $f = 18$ mm input collimator — the reverse propagation direction compared to the earlier measurement. In this configuration the $f = 25$ mm lens collimated the output beam correctly, confirming that the input beam was now properly focused, and the theoretical estimate $w_0 = 1.6$ mm was consistent with the measured collimated beam parameters. With this corrected setup, the power measured after the black fiber and the $f = 18$ mm collimator reached about 70% of the power measured before the fiber, i.e. directly after the $f = 25$ mm lens.

This result was confirmed in a repeated measurement, which gave again an injection efficiency of about 70%. At the same time, bending losses were reassessed by flexing the fiber and monitoring the power after the collimator, confirming that they remain negligible.

Planned next steps include inserting a half-wave plate before the $f = 25$ mm lens, further optimizing the injection efficiency by fine adjustment of the alignment, and installing a half-wave plate and polarizing beam splitter after the fiber to evaluate polarization purity and stability. A dedicated 1064 nm fiber with a flat (rather than angled) connector is also being considered, in order to avoid the astigmatism introduced by angled connectors.

1.3.7 From test bench to the definitive injection branch

The measurements described above were carried out on a preliminary test bench, meant to validate the injection strategy before committing to the final optical layout. Once the approach was confirmed, the actual injection branch feeding the black fiber was built downstream of the polarizing beam splitter that separates the hollow-core and conveyor-belt paths of the setup.

Figure 1.9: *é SGRANATA*

Along this branch, the beam first passes through an AOM, used for fast intensity and frequency control, followed by a steering mirror. A telescope built from an $f = -30$ mm and an $f = 150$ mm lens then re-collimates the beam to the target radius of 1.6 mm established in the previous characterization. Two further mirrors steer the collimated beam into the injection path, where a half-wave plate sets the input polarization along the slow axis of the polarization-maintaining fiber before the beam is focused into the fiber by the $f = 25$ mm lens, again mounted on a micrometric z -stage to compensate for the fiber endcap as described above.

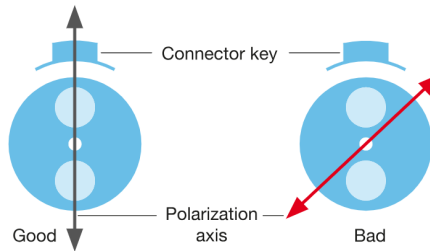


Figure 1.10

With this definitive branch in place, the injection efficiency reached 75%, consistent with the test-bench results and confirming that the injection strategy developed above transfers correctly to the full optical layout. With efficient and reproducible injection secured, the following section turns to characterizing the polarization stability of this system.

1.4 Polarization Stability Characterization

DIRE CHE A UNA CERTA VENGONO NORMALIZZATI I SEGNALI E FATTA LA DIFFERENZA

With efficient injection secured, the next step is to characterize how well the black fiber preserves the input polarization state over time, since this directly determines how stable

the optical potential derived from this beam will be.

1.4.1 Detection setup and PBS alignment

Polarization stability was assessed by placing a polarizing beam splitter (PBS) after the fiber and monitoring both of its outputs simultaneously: the transmitted beam was detected with a home-made photodiode (channel 1), and the reflected beam with a Thorlabs PDA100A2 photodiode (channel 2, operated at 0 dB of internal gain with 6 dB of additional electrical attenuation). The total power incident on the PBS was 4.2 mW, with the input polarization initially set at 45° to the PBS axes — found experimentally as the orientation that splits the power evenly between the two outputs.

Before any stability measurement, the alignment of the PBS itself was checked, since the residual power appearing in reflection when the input polarization is meant to be fully transmitted (or vice versa) turned out to be very sensitive to the fine orientation of the PBS. With careful adjustment, a polarization extinction ratio (PER) in excess of 30 dB was obtained on both outputs — 31 dB in reflection and 33 dB in transmission — confirming that the PBS itself was not a limiting factor for the subsequent measurements. Time traces of both photodiode signals were then acquired over a range of durations, always with 10^5 data points per trace: 600 ms, 60 s, and 600 s. The 60 s traces were acquired in two variants, a plain sampling mode — better suited to resolving fast, sub-millisecond noise — and a High-Resolution (HR) acquisition mode, better suited to the longer traces where the relevant fluctuations are slower.

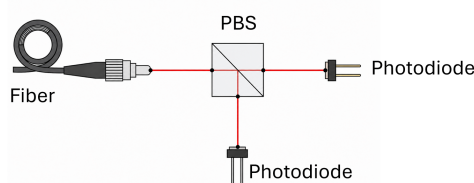


Figure 1.11

1.4.2 Calibration of the home-made photodiode

The home-made photodiode used on the transmission channel is an older unit, inherited from the LENS laboratory, built around a BPW34 photodiode. This photodiode has a peak responsivity of about 0.7 A/W near 900 nm, which drops to roughly $0.7 \times 0.3 = 0.21$ A/W at 1064 nm; the photocurrent is converted to a voltage by an OP27G operational amplifier configured as a transimpedance amplifier with a 10 k Ω gain, followed by an output stage with 1 k Ω output impedance. Taken together, these figures predict an output of about 2.1 V for 1 mW of incident power at 1064 nm, consistent with the calibration used in the measurements below.

As a cross-check on the overall injection performance at this stage, the fiber coupling efficiency was re-measured and found to be 69% (4.18 mW input, 2.88 mW output).

1.4.3 Aligning the detection to the fiber’s principal axis

To make sure the polarization analysis was actually referenced to the fiber’s own polarization axes, rather than to an arbitrary lab-frame direction, the fiber output was equipped with the output collimator, a mirror, a half-wave plate (HWP), a PBS, and a power meter placed in transmission. The idea is that when the input and output HWPs are rotated so that the transmitted power is maximized and the reflected power minimized, the light is linearly polarized and aligned with one of the PM fiber’s principal axes; this condition was reached for an input HWP angle of 131° .

Once the beam was confirmed to be linearly polarized in this way, it was used to finely re-align the PBS itself, exploiting the fact that the reflected-port signal is particularly sensitive to small rotations of the PBS. The output HWP was then rotated to maximize the transmitted power, reaching 2.68 mW. The power meter was subsequently moved to the reflection port, and the reflected signal was minimized by jointly adjusting the PBS orientation and fine-tuning both HWPs. This procedure yielded a minimum reflected power of $4\mu\text{W}$ (against a measured background of $1.5\mu\text{W}$) and a maximum of 2 mW, corresponding to a polarization extinction ratio of 27 dB in reflection.

1.4.4 Long-term stability: Allan deviation and power spectral density

For the actual stability measurements, the detection scheme was updated to use the home-made photodiode on the reflection port (channel 1) and the Thorlabs photodiode on the transmission port (channel 2, now operated at 10 dB of gain and without electrical attenuation). Two complementary tools were used to analyze the resulting time traces: the Allan deviation and the power spectral density (PSD) of the photocurrent.

The Allan deviation $\sigma_y(\tau)$ quantifies how much a signal’s average value changes between consecutive intervals of duration τ , as a function of τ itself. Unlike a simple standard deviation over the whole trace, it isolates fluctuations occurring specifically at the averaging time τ , which makes it well suited to distinguishing fast noise from slow drifts and to identifying, from the shape of $\sigma_y(\tau)$, which noise process (e.g. white noise, flicker noise, or a genuine long-term drift) dominates at each timescale — the same tool used elsewhere in this work to characterize the master-laser frequency stability, applied here instead to the polarization-sensitive optical power.

The power spectral density of each photodiode signal was computed as a complementary, frequency-resolved view of the same fluctuations: rather than looking at how the signal's average drifts with the averaging time, the PSD shows how the total noise power is distributed across Fourier frequencies, making it straightforward to spot discrete noise sources (e.g. line-frequency pickup or mechanical resonances) as narrow peaks superimposed on the broadband noise floor. To assess whether the observed fluctuations are limited by fundamental photon statistics or by excess technical noise, the measured PSD was compared against the shot-noise level expected for the same average optical power, following the derivation below.

1.4.4.1 From photocurrent noise to light-power noise

For a photodetector with quantum efficiency QE, the number of photoelectrons generated in a time interval Δt inherits its fluctuations partly from the photon-number fluctuations of the incident light and partly from the quantum efficiency itself. The variance of the electron number can be written as

$$\sigma_n^2 = (\text{QE})^2 \sigma_N^2 + \bar{n}, \quad (1.37)$$

where σ_N^2 is the variance of the incident photon number over Δt , and $\bar{n} = \text{QE} \bar{N}$ is the mean number of photoelectrons generated by a mean photon number $\bar{N}$.

Since the photocurrent is $I = ne/\Delta t$ and the optical power is $P = \hbar\omega N/\Delta t$, this translates directly into the photocurrent variance,

$$\sigma_I^2 = \left(\frac{\text{QE} e}{\Delta t} \right)^2 \sigma_N^2 + \frac{\bar{I} e}{\Delta t}, \quad (1.38)$$

so that, after subtracting the pure electron shot-noise term $\bar{I}e/\Delta t$, the excess variance is directly proportional to the variance of the light power itself,

$$\sigma_I^2 - \frac{\bar{I}e}{\Delta t} = \left(\frac{\text{QE} e}{\hbar\omega} \right)^2 \sigma_P^2. \quad (1.39)$$

The combination $\eta \equiv \text{QE} e/(\hbar\omega)$ is simply the photodiode's responsivity; for the Thorlabs photodiode at 1064 nm, $\eta = 0.4 \text{ A/W}$. The light-power variance can therefore be recovered from the measured photocurrent variance as

$$\sigma_P^2 = \frac{\sigma_I^2 - \sigma_{I,\text{SN}}^2}{\eta^2}, \quad (1.40)$$

where the light-power power spectral density is obtained from the excess of the measured photocurrent PSD over its shot-noise floor $\sigma_{I,\text{SN}}^2 = \bar{I}e/\Delta t$. The corresponding shot-noise-limited power fluctuation is

$$\sigma_{P,\text{SN}}^2 = \frac{\bar{P} \hbar \omega}{\Delta t} = 2\bar{P} \hbar \omega \Delta f, \quad (1.41)$$

which provides the reference level against which the measured power fluctuations — and hence the polarization stability of the fiber — are compared.

1.5 Numerical Model of EIT Cooling

To study Electromagnetically Induced Transparency (EIT) cooling for our system, two numerical models of increasing complexity were implemented and simulated with the QuTiP package: an idealized three-level Λ system, used to validate the cooling mechanism and its numerics in a minimal setting, and a realistic 24-level model of ^{87}Rb including the full hyperfine and Zeeman structure. Both models describe an atom coupled to a single quantized motional mode, and share the same underlying master-equation formalism; the difference lies entirely in the internal atomic structure retained in the simulation.

1.5.1 Atomic Level Structure

In the minimal model, the atom is reduced to three states, two ground states $|g_1\rangle, |g_2\rangle$ and one excited state $|e\rangle$, coupled respectively by a weak probe field (Rabi frequency Ω_p , detuning Δ_p) and a strong coupling field (Rabi frequency Ω_c , detuning Δ_c). This bare Λ system reproduces the dark-state interference exactly by construction and was used to benchmark the solvers and the overall cooling dynamics before moving to the full atomic structure.

The realistic model instead retains the complete set of hyperfine and Zeeman sublevels of ^{87}Rb involved in the D_2 transition, labeled $|F, m_F\rangle$: the ground manifolds $F = 1$ (3 states) and $F = 2$ (5 states), and the excited manifolds $F' = 0, 1, 2, 3$ (16 states), for a total atomic Hilbert space of dimension 24. Every dipole-allowed transition between these sublevels is included, weighted by the corresponding Clebsch–Gordan coefficient $C_{g,e}^{(q)} = \langle F_g, m_g; 1, q | F_e, m_e \rangle$ for the three polarizations $q \in \{-1, 0, 1\}$.

Rather than using a central sublevel such as $m_F = 0$ — from which an excited atom can decay dipole-allowed into up to six different ground sublevels, quickly scattering population out of reach of the cooling lasers — the EIT scheme was built around an edge sublevel, $m_F = -2$. Selection rules ($\Delta m_F = 0, \pm 1$) then restrict spontaneous decay from the relevant excited states to at most three ground sublevels, which allows the cooling scheme to be closed with the addition of a single repump field. Two equivalent ways of assigning the probe, coupling, and repump lasers to these three sublevels were implemented as separate configurations: one in which both Λ -system ground states lie in $F = 2$ and the repumper recovers population that has decayed to $F = 1$, and one in which the two Λ -system ground states are split between $F = 2$ and $F = 1$ and the repumper instead recovers population left behind in $F = 2$. The second configuration, closer to the level structure available in our experimental setup, is the one carried forward into the real-parameter (“axial”) simulation described below.

1.5.2 Hilbert Space and Motional Coupling

The full simulated system lives in the tensor-product space $\mathcal{H} = \mathcal{H}_{\text{atom}} \otimes \mathcal{H}_{\text{motion}}$, where $\mathcal{H}_{\text{motion}}$ is a truncated harmonic oscillator basis $\{|n\rangle\}$ describing the quantized motion along the trap axis of interest, with ladder operators a , $a^\dagger$ and number operator $n = a^\dagger a$, and trap frequency ν . The truncation of the motional basis (the maximum phonon number N_{vib} retained) was chosen large enough that the population reaching the highest included Fock state stays negligible throughout the simulated evolution, while being kept as small as computationally feasible given how steeply the size of the full problem grows with it. Motion is coupled to the internal atomic dynamics through the recoil accumulated in absorption and emission of a photon. In the Lamb–Dicke regime ($\eta \ll 1$, with η the Lamb–Dicke parameter of the relevant beam) the corresponding operator is linearized as $e^{ikx} \approx 1 + i\eta(a + a^\dagger)$, which is the approximation used throughout this work to couple the internal and motional degrees of freedom.

1.5.3 Hamiltonian

The total Hamiltonian is written as

$$H = H_{\text{free}} + H_{\text{Zeeman}} + H_{\text{int}}, \quad (1.42)$$

with a free part

$$H_{\text{free}} = \sum_i E_i |i\rangle\langle i| + \nu a^\dagger a \quad (1.43)$$

collecting the bare atomic energies and the motional energy. For the 24-level model, an external magnetic field B lifts the degeneracy of the m_F sublevels through

$$H_{\text{Zeeman}} = \sum_i g_F \mu_B B m_F |i\rangle\langle i|. \quad (1.44)$$

The atom–light interaction is split into two parts. The EIT beams (probe and coupling) explicitly carry the motional coupling introduced above,

$$H_{\text{EIT}} = \sum_{g,e,q} \frac{\Omega_{\text{EIT}}}{2} C_{g,e}^{(q)} |e\rangle\langle g| \left(1 + i\eta(a + a^\dagger)\right) + \text{h.c.}, \quad (1.45)$$

while the repump field, whose role is purely to recycle population out of the “lost” ground states g_{lost} back into the active cooling cycle rather than to participate in the cooling transition itself, is treated without motional sidebands,

$$H_{\text{repump}} = \sum_{g_{\text{lost}},e,q} \frac{\Omega_{\text{rep}}}{2} C_{g,e}^{(q)} |e\rangle\langle g_{\text{lost}}| + \text{h.c.}, \quad (1.46)$$

so that $H_{\text{int}} = H_{\text{EIT}} + H_{\text{repump}}$.

In the reduced three-level model, both the Zeeman term and the repump field are absent, and the Hamiltonian collapses to

$$H = \Delta_p |g_1\rangle\langle g_1| + \Delta_c |g_2\rangle\langle g_2| + \nu a^\dagger a + \frac{\Omega_c}{2} (\sigma_{eg_2} + \sigma_{g_2e}) + \frac{\Omega_p}{2} [\sigma_{eg_1} (1 + i\eta(a + a^\dagger)) + \text{h.c.}], \quad (1.47)$$

with $\sigma_{egi} = |e\rangle\langle g_i|$. In this one-dimensional implementation of the three-level model, the coupling beam is taken to propagate orthogonally to the motional axis (so that its effective Lamb–Dicke parameter vanishes, $\eta_c = 0$), while the probe beam propagates along the axis and carries the motional coupling; in the general 24-level Hamiltonian, motional coupling can instead be assigned to any of the beams depending on the laboratory geometry being modeled, which is how the axial-direction configuration described below was set up.

1.5.4 Dissipation and the Master Equation

The system evolves under the Lindblad master equation

$$\frac{d\rho}{dt} = -i[H, \rho] + \mathcal{L}(\rho), \quad \mathcal{L}(\rho) = \sum_k \left(L_k \rho L_k^\dagger - \frac{1}{2} \{L_k^\dagger L_k, \rho\} \right), \quad (1.48)$$

with the jump operators L_k encoding spontaneous emission. In the three-level model, decay from $|e\rangle$ is split symmetrically between the two ground states,

$$L_1 = \sqrt{\gamma/2} |g_1\rangle\langle e|, \quad L_2 = \sqrt{\gamma/2} |g_2\rangle\langle e|. \quad (1.49)$$

In the 24-level model, every dipole-allowed decay channel is included explicitly, again weighted by its Clebsch–Gordan coefficient,

$$L_{e \rightarrow g}^{(q)} = \sqrt{\gamma} C_{g,e}^{(q)} |g\rangle\langle e|, \quad (1.50)$$

acting trivially on the motional degree of freedom ($L_{e \rightarrow g}^{(q)} \rightarrow L_{e \rightarrow g}^{(q)} \otimes \mathbb{I}_{\text{motion}}$). It is this explicit multi-channel decay — allowing population to fall outside the two-state cooling subspace addressed by the probe and coupling lasers — that makes the repump field necessary in the realistic model, unlike in the closed three-level system.

1.5.5 Steady-State Spectrum

For a fixed set of laser parameters, the steady state ρ_{ss} satisfying $d\rho/dt = 0$ is computed directly, and used to evaluate the absorption spectrum as the total excited-state population,

$$A(\Delta_p) = \text{Tr}(\rho_{ss} P_e), \quad P_e = \sum_e |e\rangle\langle e|, \quad (1.51)$$

which reduces to $A(\Delta_p) = \langle e|\rho_{ss}|e\rangle$ in the three-level case. Scanning Δ_p at fixed Δ_c maps out the Fano-shaped spectrum used to identify the position of the dark resonance and of the motional sidebands, expected at $\Delta_p = \Delta_c \pm \nu$. In the 24-level model this same steady-state calculation is also used to track, alongside $A(\Delta_p)$, the population accumulated in the target excited manifold ($P_{e2} = \sum_{e \in F'=2} |e\rangle\langle e|$), in the off-resonant excited manifold ($P_{e3} = \sum_{e \in F'=3} |e\rangle\langle e|$), and in the ground sublevels outside the intended cooling cycle ($P_{\text{leak}} = \sum_{g \neq g_{\text{target}}} |g\rangle\langle g|$), which is how the laser parameters (coupling detuning, probe Rabi frequency, repump Rabi frequency) were subsequently scanned and compared against each other.

1.5.6 Time-Dependent Dynamics

The actual cooling process is followed by propagating the master equation in time and evaluating the mean phonon number

$$\langle n(t) \rangle = \text{Tr}(a^\dagger a \rho(t)) \quad (1.52)$$

starting from a specified initial motional state. Two complementary solvers were used for this purpose. For the three-level model, and for heavily truncated versions of the 24-level model, the exact Lindblad evolution was integrated directly with QuTiP's `mesolve`, which propagates the full density matrix; this is exact and noise-free but scales as $O(N^2)$ in the total Hilbert space dimension N , which becomes prohibitive for the full 24-level model once 24 internal states are combined with ~ 15 motional levels ($N = 360$, corresponding to a 360×360 density matrix).

For the complete 24-level model, the time evolution was instead computed with the Monte Carlo wavefunction method (`mcsolve`), which propagates individual state vectors $|\psi(t)\rangle$ subject to stochastic quantum jumps representing spontaneous emission events, and reconstructs the density matrix as the ensemble average

$$\rho(t) = \mathbb{E}[|\psi(t)\rangle\langle\psi(t)|] \quad (1.53)$$

over a set of independent trajectories (`ntraj`). This approach scales only as $O(N)$ in memory and parallelizes naturally across trajectories, at the cost of statistical noise in the resulting curves, which is reduced by increasing `ntraj` at the expense of computation time. For both solvers, the relevant control parameters (total simulated time, number of saved time points, and, for `mesolve`, the maximum number of internal integration steps) were adjusted so as to resolve the cooling dynamics without exceeding the memory or

runtime available; state storage was disabled for the Monte Carlo runs on the full 24-level system, since retaining the full state vector for a large number of trajectories over a 360-dimensional space is not memory-feasible.

Simulation time is expressed throughout in units of the excited-state decay rate γ , so that one simulation time unit corresponds to one excited-state lifetime; converting to physical units requires accounting for the angular-frequency convention used internally, giving a physical duration per simulation time unit of $1/(2\pi \times 6.067 \text{ MHz}) \approx 26.23 \text{ ns}$ for the ^{87}Rb D_2 line used in the realistic model.

1.5.7 Parameter Optimization Procedure

Within the 24-level model, the laser parameters were optimized sequentially using the steady-state spectrum and leakage populations described above: first the repump Rabi frequency, to identify the minimum intensity that sufficiently depletes the unwanted ground-state population without introducing an excessive AC Stark shift of the Fano profile; then the probe Rabi frequency, trading off excitation amplitude against the power broadening of the dark resonance; and finally the coupling-laser detuning, to re-align the Fano absorption maximum with the target motional sideband once the repump- and probe-induced light shifts were accounted for. Each parameter was scanned individually while holding the others fixed, using the same steady-state diagnostics (Fano profile, ground-state leakage, and $F' = 3$ off-resonant population) for every value tested; the resulting optimized parameter set was then carried into the time-dependent Monte Carlo simulation of the full cooling dynamics.

1.5.8 Axial-Direction Simulation with Real System Parameters

As a final step, the configuration with the two Λ -system ground states split across $F = 1$ and $F = 2$ was re-run using parameters specific to our experimental setup rather than the generic values used in the earlier optimization stages. In particular, the trap frequency ν and the Lamb–Dicke parameter η entering the motional coupling were set to the values corresponding to the axial direction of our trap geometry, in place of the illustrative values used for the parameter scans described above. The same optimization procedure — steady-state Fano spectrum, ground-state leakage tracking, and sequential adjustment of the coupling detuning, probe Rabi frequency, and repump Rabi frequency — was applied to this axial configuration, and the resulting parameter set was then used to run the full Monte Carlo time evolution of the mean phonon number for this specific geometry, following the same procedure described in the previous subsection.

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